

Polytropic Index Derivation

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Background

When thinking about compressing or expanding air in a piston, there are three ways in which the process can occur: adiabatically, isothermally, or realistically (which is somewhere in between adiabatic and isothermal). In the adiabatic case, there is no heat transfer involving the gas in the piston, so the gas in the piston experiences a change in both temperature and pressure. In the isothermal case, there is infinite heat transfer, so the gas in the piston experiences only a change in pressure. Finally, in the realistic case, there is finite heat transfer so the gas will experience some change in temperature and a change in pressure. In order to increase round trip efficiency as much as possible, our goal is to get as isothermal as possible. We'll only ever approach that case, and of course we'll always incur some change in temperature of the gas in the piston since infinite heat transfer is not possible. In order to quantify how much heat transfer there is, the following polytropic equations (which assume an ideal gas) are often used:

$$PV^n = \text{constant} \quad \text{Eq. 1}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^n \quad \text{Eq. 2}$$

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{n-1} \quad \text{Eq. 3}$$

Here, P is pressure, V is volume, T is temperature, and n is the polytropic index, where $n = 1$ denotes the isothermal case, $n = \gamma = C_p/C_v$ denotes the adiabatic case (for air $\gamma = 1.4$), and $1 < n < \gamma$ denotes the realistic case. In order to try to make more efficient pistons, it is necessary to be able to predict the realistic polytropic index, which will change based on the heat transfer available in the system and the system parameters. Therefore, the goal of this derivation is to find an expression to predict the polytropic index of a piston (which is used to find its efficiency) based on only the system parameters, such as volume ratio, piston size, etc.

Iterative Solution

Polytropic Index

The goal of the iterative solution section is to find a method to predict the polytropic index of a process that is as accurate as possible. Thus, simplicity will be sacrificed for accuracy, and calculating the polytropic index this way will be very involved. However, it should result in (hopefully) an accurate polytropic index estimate, and therefore an accurate efficiency estimate.

For a compression, each scenario (adiabatic, isothermal, or realistic) involves adding energy into the system through compressing the gas, and this energy flows into different places based on what scenario is occurring. In the adiabatic case it is stored as a temperature change in the gas, in the isothermal case it flows out of the gas through heat transfer, and in the realistic case some of the energy goes into a temperature change in the gas and some of the energy goes into heat transfer. However, because different n -values involve different temperature changes in the gas, and thus reach different peak pressures, the work done on the gas for different compressions is a function of n , the polytropic index. All of this is true for expansion as well, but the energy flow is in the opposite direction. Thinking in terms of a compression, we can set this work done on the gas equal to the change in internal energy in the gas and the heat that flows out of the gas:

$$W_{in} = \int_{V_f}^{V_0} P dV = \Delta U_{gas} + Q \quad \text{Eq. 4}$$

Here P is pressure, V_f is the final volume, V_0 is the initial volume, U_{gas} is the internal energy of the gas, and Q is heat that's flowed out of or into the system. Using the polytropic equations we can substitute a function of volume for the pressure term:

$$W_{in} = \int_{V_f}^{V_0} P_0 \left(\frac{V_0}{V} \right)^n dV \quad \text{Eq. 5}$$

where V is the instantaneous volume of the gas. Solving this integral for the limit of $n > 1$, we get

$$W_{in} = P_0 V_0^n \frac{(V_0^{1-n} - V_f^{1-n})}{1-n} \quad \text{Eq. 6}$$

This expression can be simplified to the following:

$$W_{in} = P_0 V_0 \frac{(VR^{n-1} - 1)}{n-1} \quad \text{Eq. 7}$$

where P_0 is the initial pressure and VR is the volume ratio, where

$$VR = \frac{V_0}{V_f} \quad \text{Eq. 8}$$

We can then plug Eq. 7 back into Eq. 4 to get

$$W_{in} = P_0 V_0 \frac{(VR^{n-1} - 1)}{n-1} = \Delta U_{gas} + Q \quad \text{Eq. 9}$$

Since the gas in the piston is likely air, which is composed of mainly diatomic molecules, we can model the change in internal energy of the gas in the piston with the following equation:

$$\Delta U_{gas} = \frac{5}{2} k_B N_p (T_f - T_0) \quad \text{Eq. 10}$$

where k_B is the Boltzmann constant, N_p is the number of molecules in the air, T_f is the final temperature of the air after the process (compression or expansion), and T_0 is the initial temperature of the gas. Another way to think about this is by rewriting Eq. 10 as the following:

$$\Delta U_{gas} = C_v m_{gas} (T_f - T_0) \quad \text{Eq. 11}$$

where C_v is the heat capacity for constant volume of the gas. If we assume that instead of air the gas in the piston is an ideal gas, the internal energy change can be written as the following

$$\Delta U_{ideal\ gas} = \frac{a_{moles} R_{gas}}{\gamma - 1} (T_f - T_0) \quad \text{Eq. 12}$$

where a_{moles} is the number of moles in the gas, R_{gas} is the gas constant, and γ is the ratio of C_p (heat capacity for constant pressure) to C_v for the gas. This can be rewritten again as

$$\Delta U_{ideal\ gas} = \frac{1}{\gamma-1} k_B N_p (T_f - T_0) \quad \text{Eq. 13}$$

For air, $\gamma = 1.4$, so plugging this in simplifies the equation to the following

$$\Delta U_{ideal\ gas} = \frac{1}{1.4-1} k_B N_p (T_f - T_0) = \frac{5}{2} k_B N_p (T_f - T_0) \quad \text{Eq. 14}$$

which is the same as before! So the two expressions for ΔU_{gas} are identical. I'll use the following rewritten version as a general case

$$\Delta U_{gas} = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (T_f - T_0) \quad \text{Eq. 15}$$

where M_{gas} is the molar mass of the gas in the piston. This version involves the most convenient form so I'll use this one from now on.

To get the expression for Q_{real} , we have to integrate the heat transfer rate over the process (either a compression or an expansion). In a general sense, steady state convection and conduction can be modeled as

$$\frac{dQ}{dt} = \frac{T_g - T_{other}}{R} \quad \text{Eq. 16}$$

Where T_g is the instantaneous temperature of the gas in the piston, T_{other} is the temperature of the substance the gas is transferring heat to, and R is the thermal resistance to the flow of heat out of the gas. Assuming $T_{other} = T_0$, or the ambient air temperature (which I'll also assume is the starting temperature before the process begins), we must find an expression for T_g as a function of time. Since we can relate the instantaneous gas temperature to the volume ratio via the polytropic equations, we can model the instantaneous temperature of the gas as

$$T_g(t) = T_0 \left(\frac{V_0}{V(t)} \right)^{n-1} \quad \text{Eq. 17}$$

The next step requires that we know how volume varies with time in our system. Nominally, it will vary sinusoidally, but one other common way it might vary is linearly. These would yield the following two functions for $V(t)$:

Sinusoidal

$$V(t) = \frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2} \quad \text{Eq. 18}$$

Linear

$$V(t) = -\frac{V_0 - V_f}{t_p} t + V_0 \quad \text{Eq. 19}$$

Where both of these equations result in an initial volume of V_0 , a final volume of V_f , and a process time of t_p . Plugging these back into Eq. 17, we get

$$T_g(t) = T_0 \left(\frac{V_0}{\frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2}} \right)^{n-1} \quad \text{Eq. 20}$$

$$T_g(t) = T_0 \left(\frac{V_0}{-\frac{V_0 - V_f}{t_p} t + V_0} \right)^{n-1} \quad \text{Eq. 21}$$

In order to visualize these functions, we can plot both of them for an arbitrary VR , n , t_p and T_0 value and compare them. This is shown below in Fig. 1.

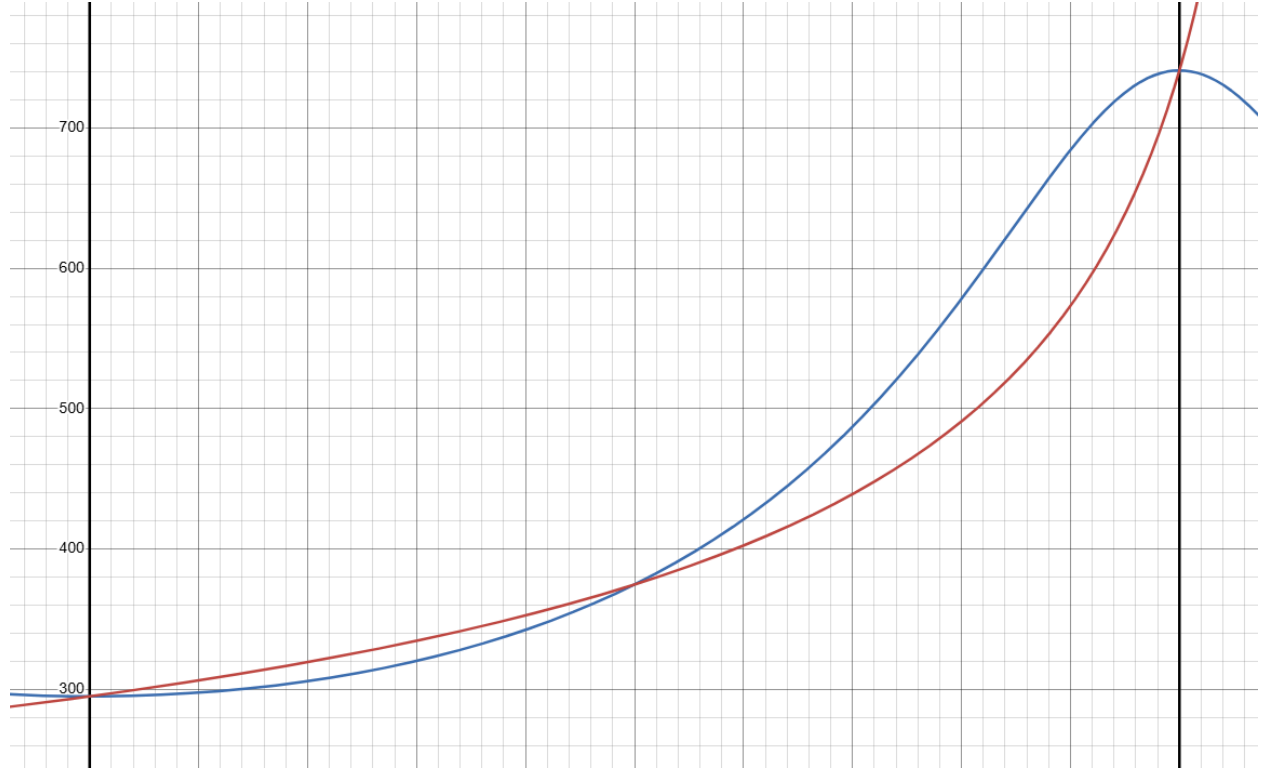


Fig. 1. Temperature of an example sinusoidal (blue) vs linear (red) compression vs time. Note the vertical bold black lines indicate t_0 (left) and t_p (right), and that the arbitrary parameters here (VR , T_0 , etc) are the same for the two functions.

Plugging these into Eq. 16 and integrating, we get

$$Q = \int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{\frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2}} \right)^{n-1} - T_0 \right) \right) dt \quad \text{Eq. 22}$$

$$Q = \int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{-\frac{V_0 - V_f}{t_p} t + V_0} \right)^{n-1} - T_0 \right) \right) dt \quad \text{Eq. 23}$$

Both of these are very difficult integrals to compute. For simplicity, it would be nice if we could approximate the temperature as the following

$$T(t) = T_0 + (T_f - T_0) \left(\frac{t}{t_p} \right)^b \quad \text{Eq. 24}$$

Where b is some constant. This would give us

$$Q = \int_0^{t_p} \left(\frac{1}{R} (T_0 + (T_f - T_0) \left(\frac{t}{t_p} \right)^b - T_0) \right) dt = \frac{(T_f - T_0)t_p}{(b+1)R} \quad \text{Eq. 25}$$

Which is equivalent to the same expression but with a different constant, W , as seen below

$$Q = \frac{W(T_f - T_0)t_p}{R} \quad \text{Eq. 26}$$

This expression is simple and contains important system parameters t_p and R . It also contains $(T_f - T_0)$ just like Eq. 15, which is convenient because it will allow us to solve for n more easily later. So, setting the sinusoidal compression heat exchange expression (Eq. 22) equal to our assumed heat exchange expression (Eq. 26), what does the factor W need to be to make these two equal? Unfortunately, Eq. 22 does not have an analytical solution, so that can only be found with the following:

$$\frac{W(T_f - T_0)t_p}{R} = \int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{\frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2}} \right)^{n-1} - T_0 \right) \right) dt \quad \text{Eq. 27}$$

$$W = \frac{R}{(T_f - T_0)t_p} \int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{\frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2}} \right)^{n-1} - T_0 \right) \right) dt \quad \text{Eq. 28}$$

$$W = \frac{1}{(VR^{n-1} - 1)t_p} \int_0^{t_p} \left(\left(\frac{V_0}{\frac{V_0 - V_f}{2} \cos\left(\frac{\pi t}{t_p}\right) + \frac{V_0 + V_f}{2}} \right)^{n-1} - 1 \right) dt \quad \text{Eq. 29}$$

This final equation can be evaluated with numerical methods or a definite integral solver. For the linear solution, we start by equating Eq. 23 to Eq. 26 giving us

$$\frac{W(T_f - T_0)t_p}{R} = \int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{-\frac{V_0 - V_f}{t_p} t + V_0} \right)^{n-1} - T_0 \right) \right) dt \quad \text{Eq. 30}$$

Fortunately, the RHS has an analytical solution, which is found as follows:

$$\int_0^{t_p} \left(\frac{1}{R} \left(T_0 \left(\frac{V_0}{-\frac{V_0-V_f}{t_p}t + V_0} \right)^{n-1} - T_0 \right) \right) dt = \frac{T_0 t_p}{R} \left(\frac{2V_f - V_0 - V_f V R^{n-1} - n(V_f - V_0)}{(V_f - V_0)(n-2)} \right) \quad \text{Eq. 31}$$

Next we can substitute this back into Eq. 30 and solve for W .

$$\frac{W(T_f - T_0)t_p}{R} = \frac{T_0 t_p}{R} \left(\frac{2V_f - V_0 - V_f V R^{n-1} - n(V_f - V_0)}{(V_f - V_0)(n-2)} \right) \quad \text{Eq. 32}$$

$$W = T_0 \left(\frac{2V_f - V_0 - V_f V R^{n-1} - n(V_f - V_0)}{(V_f - V_0)(n-2)(T_f - T_0)} \right) \quad \text{Eq. 33}$$

We can use the polytropic equations to simplify this expression to

$$W = \frac{2V_f - V_0 - V_f V R^{n-1} - n(V_f - V_0)}{(V_f - V_0)(n-2)(V R^{n-1} - 1)} \quad \text{Eq. 34}$$

This gives us an exact expression for the factor W so that we can now represent the total heat transfer Q with Eq. 26. Notice that Eq. 34 for W contains n , yet the whole point of this analysis is to predict n . We could estimate n as the same constant for all processes in Eq. 34 (for example, $n = 1.2$, the average between the adiabatic and isothermal case), but this will lead to a less robust and less accurate theory. We will see later that an initial n value is necessary to calculate other parts of the theory as well, so in the end we will have to iterate for an n value based on some initial guess. If your system has a sinusoidal volume profile but you don't have a numerical integral solver handy, you could use the linear solution as an approximation. The accuracy of this approximation will depend on VR and n , but in general the larger VR and n are the larger the error will be. For example, I found that a $VR = 5$ and an $n = 1.05$ gives an 8.1% error, but a $VR = 30$ and an $n = 1.3$ gives a 23.6% error from the true heat transfer approximating a sinusoidal compression as a linear one.

To compare the approximated temperature function with the real ones, below the sinusoidal, linear, and approximated temperatures are plotted after solving for W with the same arbitrary scenario as before.

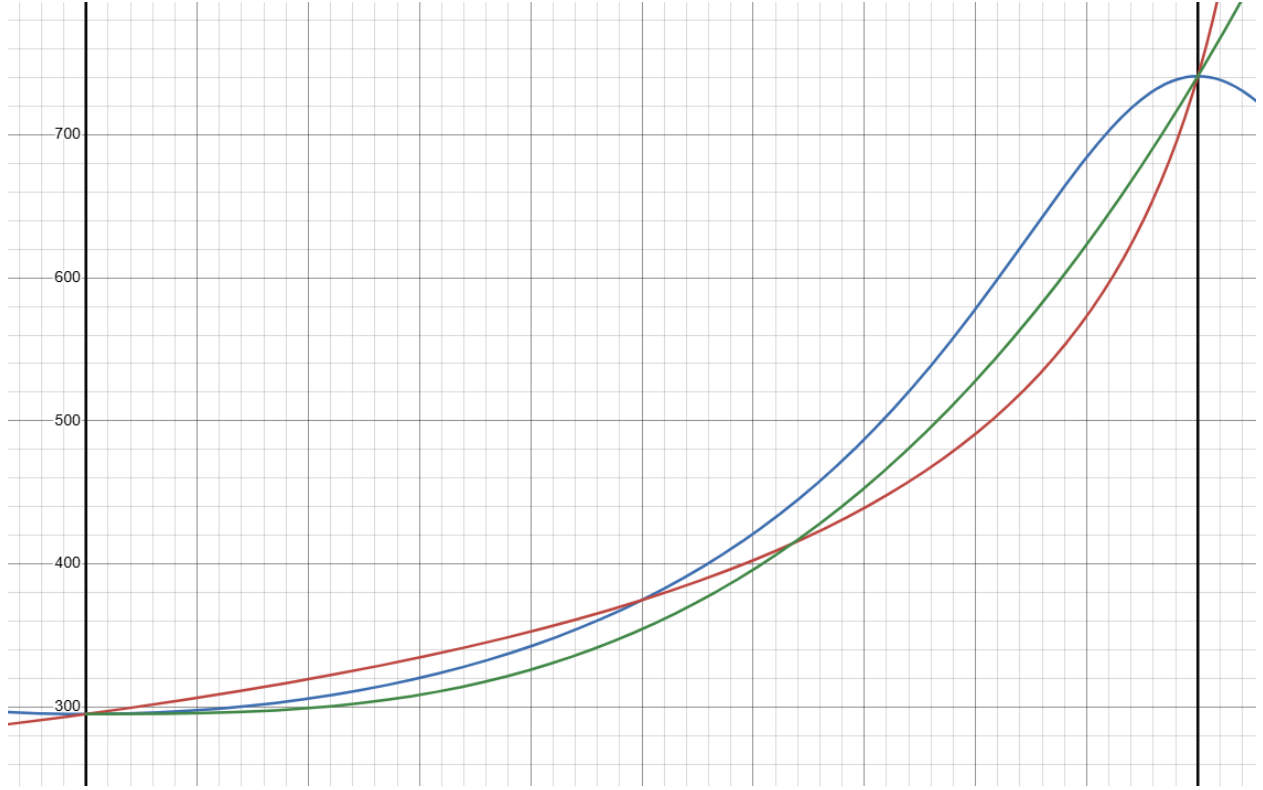


Fig. 2. Temperature of an example sinusoidal (blue), linear (red), and approximated (green) compression vs time. Notice that the approximated temperature function does not match the linear temperature function, but rather the area underneath the approximated temperature function matches the area underneath the linear temperature function. Further notice that both of these have a different area than the sinusoidal function.

As a reminder, the final form of Q that I'll use in the following equations is exactly correct, it's just represented in a convenient way.

Next, we can plug in all the expressions we've found into Eq. 4, and we get the following

$$P_0 V_0 \frac{(VR^{n-1}-1)}{n-1} = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (T_f - T_0) + \frac{W(n)t_p}{R} (T_f - T_0) \quad \text{Eq. 35}$$

Factoring out the temperature difference on the right side yields the following equation. I have also rewritten W as $W(n)$ to remind us that W is a function of n .

$$P_0 V_0 \frac{(VR^{n-1}-1)}{n-1} = (T_f - T_0) \left(\frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{W(n)t_p}{R} \right) \quad \text{Eq. 36}$$

We can then substitute T_f using the polytropic equations. This gives us the following

$$P_0 V_0 \frac{(VR^{n-1} - 1)}{n-1} = (T_0 VR^{n-1} - T_0) \left(\frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{W(n)t_p}{R} \right) \quad \text{Eq. 37}$$

It is impossible to solve for n analytically. So instead, we'll plug in n_0 to some of the n terms in the equation:

$$P_0 V_0 \frac{(VR^{n_0-1} - 1)}{n_0-1} = (T_0 VR^{n-1} - T_0) \left(\frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{W(n_0)t_p}{R} \right) \quad \text{Eq. 38}$$

n is now easy to solve for, and can be found by the following:

$$n = \log_{VR} \left(\frac{P_0 V_0 (VR^{n_0-1} - 1)}{T_0 (n_0-1) \left(\frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{W(n_0)t_p}{R} \right)} + 1 \right) + 1 \quad \text{Eq. 39}$$

This can be written in a few different ways, one of which is the following:

$$n = \log_{VR} \left(\frac{P_0 V_0}{T_0 \left(\frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{W(n_0)t_p}{R} \right)} \frac{(VR^{n_0-1} - 1)}{(n_0-1)} + 1 \right) + 1 \quad \text{Eq. 40}$$

Finally, it is approximately equal to

$$n = \log_{VR} \left(\left(\frac{\gamma-1}{1+W(n_0)S} \right) \left(\frac{VR^{n_0-1} - 1}{n_0-1} \right) + 1 \right) + 1 \quad \text{Eq. 41}$$

Where $\frac{\gamma-1}{1+W(n_0)S}$ and $\frac{VR^{n_0-1} - 1}{n_0-1}$ are both dimensionless, and S is the Simpson-Qin number, which represents the ratio of the process time, t_p , to the thermal timescale t_t , where

$$t_t = C_{v, gas} R = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} R \quad \text{Eq. 42}$$

In order to get an answer for n , we'll treat n_0 as an initial guess of n . After plugging in our initial guess n_0 and estimating n , we'll update our initial guess as $n_0 = n$, and we'll continue this process through iteration until we converge on a value for n .

In order to check whether or not this makes sense, and to gain some insights on how to make systems more efficient assuming Eq. 41 is correct, we can consider how the different system parameters will affect the predicted polytropic index. When looking at the VR , we can see that n increases as VR increases. This means that the more we want to compress the air (which also determines how much energy we can store) the less efficient the compressed storage will be. Next is the process time, t_p , and n gets smaller as t_p gets larger. This makes perfect sense as the slower the process is the more time there is for heat transfer. Finally, as R gets larger n increases. This must be the case as the more heat transfer there is, the smaller R will be, and the smaller n will be. Thus, the more obvious relationships that must exist are present in Eq. 41, and a few other insights can be gained as well.

We can do a sanity check with the adiabatic and isothermal cases to make sure our equation in those limits. For the adiabatic case, we can plug in the $R \rightarrow \infty$ for zero heat transfer. We should get that $n = 1.4$, and this is what our solution provides; when $R \rightarrow \infty$, $n \rightarrow 1.4$. Next we can look at the isothermal case, where $R \rightarrow 0$, or the heat exchange goes to infinity. We should get that $n = 1$, and after plugging in $R = 0$ we immediately get that $n = 1$, so this checks out as well.

Thermal Resistance

Next we have to find R , the resistance to heat transfer out of the air in the piston. This analysis will typically consider heat exchanger elements that are encasing the air in the piston (the walls and plates) but not interspersed throughout the air (like a liquid spray or porous material). In theory, this approach should work for any type of heat exchanger whether interspersed throughout the air or not, so long as the assumptions made are still valid.

Since there will be mixing of the air inside the piston due to natural convection and some bulk velocity due to the one sided compression, we can assume that the air in the piston is generally of uniform temperature. It might seem to make sense to try to find the heat transfer of the air in the piston directly to the outside air through the walls and plates of the system. This is a classic and easy heat transfer problem to solve, and is very useful when figuring out things like insulation for housing. However, this method (at least when using steady state assumptions) typically relies on the walls and plates

encasing the piston to heat up a to a temperature around $\frac{(T_f + T_0)}{2}$ on average. Since solids are so much denser than gases, they tend to require much more heat to change temperature, and this can be quantified using the thermal capacity loading, or the TCL :

$$TCL = \frac{m_{HX} c_{hx}}{m_{gas} c_{v,gas}} \quad \text{Eq. 43 (From Simpson Theory)}$$

which denotes the ratio of energy needed per degree of temperature change to heat up the heat exchanger elements and the air in the piston. If the $TCL \gg 1$, the thermal capacity in the heat exchanger elements in the piston is much larger than that of the air in the piston. Subsequently in this case the energy contained in some change of temperature in the gas in the piston would correspond to a negligible temperature change in the heat exchanger elements. Therefore, if $TCL \gg 1$, we can assume that the heat exchanger elements (walls and plates in the piston) will stay at their initial temperature even after heat transfer has occurred between the gas and the other elements in the piston, as the gas will give or take heat from the heat exchanger elements but the amount of heat is so small that it doesn't significantly affect the temperature of the heat transfer elements. This assumption also requires that the heat exchanger elements can conduct heat within themselves quickly enough to dissipate any thermal gradients that arise. This is a common assumption called the lumped system model, and is dependent on the Biot number:

$$Bi = \frac{hL_c}{k} = \frac{R_{conduction}}{R_{convection}} \quad \text{Eq. 44}$$

Where h is the convection coefficient, k is local conduction, and L_c is the characteristic distance found by dividing the heat exchanger's volume by its surface area exposed to convection. The Biot number is also equal to the ratio of the conductive thermal resistance to the convective thermal resistance. Typically, if $Bi < 0.1$, the lumped system model assumption is considered valid.

To calculate the TCL I plugged in the values for our experimental system to Eq. 20 and got that for the both plates and for the walls the $TCL \approx 300$. In an example real world system with a 4m diameter, 4m tall piston with 5cm wall thickness, you get a $TCL \approx 200$. Both systems pass the TCL criterion, thus I will assume that any increase in temperature in the walls and plates encasing the air in the piston is negligible, and subsequently that any heat transfer from the gas through the piston structure to the outside air is also negligible. The next criterion is the lumped system criterion for the walls and plates. The typical criterion states that the conductive heat transfer should be

10x the convective heat transfer, but I'll use this approach as an approximation as long as the Biot number is close to or less than 1 (convection and conduction are of the same order of magnitude).

Finally, I'll also assume that the walls and plates begin the process at ambient air temperature, T_0 , and stay at that temperature throughout the process. Thus, the only relevant heat transfer in the system is heat that flows from the gas in the piston into the walls and plates of the system (or vice versa), and this heat transfer does not meaningfully change the temperature of the walls and plates of our system.

Next we have to model the heat transfer between the gas in the piston and its surroundings. We can model this using convection, which then leads us to some important questions: which is more dominant, natural or forced convection? Are they both important? In order to figure this out, we have to look at three non-dimensional numbers: the Grashof number, the Reynolds number, and the Richardson number. The Grashof number determines the flow regime based on natural convection, the Reynolds number determines the flow regime based on fluid bulk velocity, and the Richardson number determines how dominant natural or forced convection is. If the Richardson number is much greater than one ($Ri \gg 1$) forced convection can be ignored; similarly if $Ri \ll 1$ then natural convection can be ignored. Finally, if $Ri \sim 1$, both effects must be considered.

$$Gr = \frac{g\beta(T_g - T_s)\delta^3}{\nu^2} = \frac{g(T_g - T_s)\delta^3}{T_g \nu^2} \quad \text{Eq. 45}$$

$$Re = \frac{\rho\delta u}{\mu} = \frac{\delta u}{\nu} \quad \text{Eq. 46}$$

$$Ri = \frac{Gr}{Re^2} \quad \text{Eq. 47}$$

Where g is the gravitational constant $g = 9.81 \text{ m/s}^2$, β is the fluid's coefficient of volume expansion (which can be approximated as $\beta = 1/T_g$ for an ideal gas), T_g is the instantaneous temperature of the gas, T_s is the instantaneous temperature of the surface in contact with the gas, δ is the characteristic length, ν is the kinematic viscosity of the fluid, ρ is the density of the fluid, u is the velocity of the fluid, and μ is the dynamic viscosity of the fluid. Since lots of these parameters change throughout a compression

or expansion, I'll be using the averages of them throughout the process to determine the average Grashof, Reynolds, and Richardson numbers as defined below. I'll also be defining the characteristic length scale as the diameter of the piston, D , which stays constant throughout compression and expansion.

$$\overline{Gr}_D = \frac{g(\overline{T}_g - T_s)D^3}{\overline{T}_g^{-2} \nu} \quad \text{Eq. 48}$$

$$\overline{Re}_D = \frac{D\overline{u}}{\nu} \quad \text{Eq. 49}$$

$$\overline{Ri}_D = \frac{\overline{Gr}_D}{\overline{Re}_D^2} \quad \text{Eq. 50}$$

Where an overline above a parameter denotes the average of that parameter across the relevant process (a compression or an expansion) and where the subscript D denotes the diameter based definition of a non-dimensional number. In order to calculate this, I modeled the instantaneous gas temperature as an average temperature over the whole process, where

$$\overline{T}_g = \frac{1}{t_p} \int_0^{t_p} T_g(t) dt \quad \text{Eq. 51}$$

Here we can plug in the respective temperature function that was used for the experiment. For ours we used a sinusoidal compression, so we will plug in our sinusoidal temperature profile. We will have to use our guess for n to evaluate the integral. Since we are assuming the solid components of the piston remain at $T_{\text{ambient}} = T_0 \approx 295 \text{ K}$, $T_s \approx 295 \text{ K}$. For some of the following estimates, we'll need an expression for the final length of the piston.

$$L_f = \frac{L_0}{VR} \quad \text{Eq. 52}$$

$$\overline{L} = \frac{L_0 + L_f}{2} = \frac{L_0 + \frac{L_0}{VR}}{2} = \frac{L_0}{2} \left(1 + \frac{1}{VR}\right) \quad \text{Eq. 53}$$

Where L_0 is the initial length and L_f is the final length of the air in the piston. I found the average velocity during a compression using

$$\bar{u} = \frac{x_0 - x_f}{t_p} = \frac{\frac{L_0}{2} - \frac{L_f}{2}}{t_p} = \frac{\frac{L_0}{2} - \frac{L_0}{2VR}}{t_p} = \frac{L_0}{2t_p} \left(1 - \frac{1}{VR}\right) \quad \text{Eq. 54}$$

Where x_0 and x_f denote the initial and final location of the center of mass of the gas in the piston. I found the average density during a compression as

$$\bar{\rho} = \frac{\rho_0 + \rho_f}{2} = \frac{\rho_0 + VR \rho_0}{2} = \frac{\rho_0}{2} (1 + VR) \quad \text{Eq. 55}$$

And subsequently

$$\bar{v} = \frac{\mu}{\bar{\rho}} \quad \text{Eq. 56}$$

Where μ is the dynamic viscosity of air at STP. For our typical experimental set up, $D = 0.0889 \text{ m}$, $VR \sim 3$, $L_0 = 0.4572 \text{ m}$, $\rho_0 = 1.205 \text{ kg/m}^3$, $\mu = 1.813 \cdot 10^{-5} \text{ Ns/m}^2$, and $t_p = 4 \text{ s}$. Plugging these in to get the above values, we get $\bar{\rho} = 2.41 \text{ kg/m}^3$, $\bar{u} = 0.0381 \text{ m/s}$, and $\bar{v} = 7.523 \cdot 10^{-6} \text{ Js/kg}$. Plugging all of this into the formula for the average Richardson number, I created the table below.

Table I. Grashof number, Reynolds number, and Richardson number for different polytropic index values.

n	$\overline{Gr}_D$	$\overline{Re}_D$	$\overline{Ri}_D$
1.01	$5.8 \cdot 10^5$		2.9
1.05	$2.9 \cdot 10^6$		14
1.10	$5.7 \cdot 10^6$		28

1.15	$8.6 \cdot 10^6$	450	42
1.20	$1.1 \cdot 10^7$		56
1.25	$1.4 \cdot 10^7$		70
1.30	$1.7 \cdot 10^7$		83
1.35	$2.0 \cdot 10^7$		97
1.39	$2.2 \cdot 10^7$		107

For the $n = 1.01$ case, forced convection starts to become more impactful, but natural convection is still the most important regime. For all other cases, natural convection is the dominant regime.

Next, assuming the piston only has walls (cylindrical shell), a top plate, and a piston head / bottom plate, we can assume the heat transfer looks like Eq. 16 where we want to find the total thermal resistance. Our system involves thermal resistance in parallel, so we can find the total thermal resistance as follows:

$$R = R_{total} = \left(\frac{1}{R_{walls}} + \frac{1}{R_{top\ plate}} + \frac{1}{R_{bottom\ plate}} \right)^{-1} \quad \text{Eq. 57}$$

where

$$R_i = \frac{1}{h_i A_i} \quad \text{Eq. 58}$$

Here i denotes the part of the piston you're looking at (walls, top plate, or bottom plate) A_i is the surface area of that part, and h_i is the convection coefficient for that part (i.e., a coefficient that captures the effects of the relevant convection regime for that part in that system). Since both types of convection might be important for different systems, we need to know the effects of both types of convection. Equations to find the heat transfer coefficient of natural and forced convection heat transfer in terms of the Nusselt number have been discovered empirically, and are typically defined as follows:

Free Convection

$$\overline{Nu}_D = \frac{h_n D}{k} = f(Gr, Pr) \quad \text{Eq. 59}$$

Forced Convection

$$\overline{Nu}_D = \frac{h_f D}{k} = f(Re, Pr) \quad \text{Eq. 60}$$

where Ra is the Rayleigh number, which is simply the product of the Grashof number and the Prandtl number; Ra^* is the modified Rayleigh number, which is the Rayleigh number divided by the cylinder's aspect ratio equal to L/D , Nu is the Nusselt number, the ratio of total heat transfer to conductive heat transfer in the fluid; D is the diameter of the piston, Pr is the Prandtl number, which denotes the ratio of momentum diffusivity and thermal diffusivity and can be treated as a constant for air. The piston's internal geometry is very important to the amount of free convection that occurs. Different correlations for the Nusselt number have been found for different geometries, and a few correlations are listed below:

Vertical Cylinder

$$\overline{Nu}_D = 0.39(\overline{Gr}_D)^{0.53}(Pr)^{0.7}\left(\frac{L}{D}\right)^{-1.5} \quad \text{Eq. 61}$$

Vertical Plate

$$\overline{Nu}_D = 0.67(\overline{Ra}_D)^{0.25} \quad \text{Eq. 62}$$

For the top of a horizontal plate:

$$\overline{Nu}_D = 0.54(\overline{Ra}_D)^{\frac{1}{4}} \text{ for } 10^4 < \overline{Ra}_D < 10^7 \quad \text{Eq. 63}$$

$$\overline{Nu}_D = 0.15(\overline{Ra}_D)^{\frac{1}{3}} \text{ for } 10^7 < \overline{Ra}_D < 10^{11} \quad \text{Eq. 64}$$

For the bottom of a horizontal plate:

$$\overline{Nu}_D = 0.27(\overline{Ra}_D)^{\frac{1}{4}} \text{ for } 10^5 < \overline{Ra}_D < 10^{11} \quad \text{Eq. 65}$$

For a flat surface:

$$\overline{Nu}_D = 0.664(\overline{Re}_D)^{\frac{1}{2}}(Pr)^{\frac{1}{3}} \text{ for } \overline{Re}_D < 5 * 10^5 \quad \text{Eq. 66}$$

$$\overline{Nu}_D = 0.037(\overline{Re}_D)^{\frac{4}{5}}(Pr)^{\frac{1}{3}} \text{ for } 5 * 10^5 < \overline{Re}_D < 10^7 \quad \text{Eq. 67}$$

If the piston you examine has a very low aspect ratio (i.e., as $L/D \rightarrow 0$) the relative curvature of the piston walls approaches zero and the walls can be approximated as a flat vertical plate. However, as the length of the piston walls engaged in heat transfer changes throughout the compression or expansion, this effect varies on the specific volume ratio and size of your piston.

As stated before, the dominant convection mode (free or forced) depends on the Richardson number, so we have to calculate that and then choose a convection regime that makes sense. For our setup, natural convection is pretty much dominant regardless of the polytropic index, so we can ignore forced convection. However for other systems, you'll have to calculate the Richardson number every time you want to model the heat transfer. If it turns out that the Richardson number of your system is near 1 and natural convection and forced convection are both important, one important insight that arises from this analysis is that the direction of compression and expansion matters! The typical orientation of a piston tends to involve the air being compressed upwards and expanded downwards, and this is counterproductive to maximum heat transfer as natural convection will create airflow downwards upon compression and airflow upwards upon expansion. Therefore, since the compression causes upwards velocity, and expansion causes downwards velocity, forced and free convection in this configuration always act in opposite directions. Thus I suggest that the piston direction is chosen so that it compresses the air downwards and expands the air upwards, resulting in enhanced heat transfer, with the effects of natural and forced convection working together.

If other heat exchangers are added to the system, e.g. a mesh or water droplets, you'll have to include their influence in the total thermal resistance as follows

$$R = R_{total} = \left(\frac{1}{R_{walls}} + \frac{1}{R_{top\ plate}} + \frac{1}{R_{bottom\ plate}} + \frac{1}{R_{mesh}} + \dots + \frac{1}{R_{other}} \right)^{-1} \quad \text{Eq. 68}$$

Where you'll include these individual thermal resistances in the equation if they exist in your system. I would love to include how to calculate the effects of a wire mesh or other porous media, however experimental correlations don't really exist for these implements as of right now.

Final Process

Here is the final process for calculating the polytropic index based on this theory. it will go as follows:

1. guess a value for the polytropic index, n_0
2. plug n_0 value into Eq. 41 to find the average $\frac{T_g - T_0}{T_g}$
3. Use this average $\frac{T_g - T_0}{T_g}$ to find the Richardson number, and decide which regime of convection you're in
4. Use this regime to decide which Nusselt number correlation to use, and use this to find the heat transfer coefficients for all the components of your system
5. Convert these heat transfer coefficients into a thermal resistance, and plug this into the equation for n
6. Set $n_0 = n$ and repeat steps 2-7 until $\frac{|n_0 - n|}{n} < E$ where E is an acceptable error, e.g. 10^{-6}

With these steps, you can calculate the polytropic index based on your piston's parameters.

Simplified Solution

Polytropic Index

The goal of the simplified solution section is to make a simplified version of the above theory, one that requires fewer complex calculations and no iteration. Since the goal of all of this is to figure out how efficient pistons will be, this section will focus on the following yes or no question: will a piston with known parameters be efficient? This section will utilize more approximations so it won't be as accurate, but it will be able to answer that question.

The problem set up is the same as before, with work done being converted to heat transfer and a temperature change in the gas.

$$W = \Delta U_{gas} + Q \quad \text{Eq. 69}$$

$$W = \int_{V_f}^{V_0} P dV \quad \text{Eq. 70}$$

$$W = \int_{V_f}^{V_0} P_0 \left(\frac{V_0}{V}\right)^n dV \quad \text{Eq. 71}$$

however this time, since we are only interested in efficient cases, we will assume the work done is evaluated at the isothermal limit, $n = 1$.

$$W = \int_{V_f}^{V_0} P_0 \left(\frac{V_0}{V}\right) dV = P_0 V_0 \ln(VR) \quad \text{Eq. 72}$$

Note that the magnitude of work is the same for expansion or compression, the sign just changes. Conveniently, since $n = 1$, the expression for work done is now independent of the polytropic index. Next, the internal energy change of the gas will be the same expression as before. so

$$\Delta U_{gas} = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (T_f - T_0) \quad \text{Eq. 73}$$

So far we have

$$P_0 V_0 \ln(VR) = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (T_f - T_0) + Q \quad \text{Eq. 74}$$

For the heat transfer term, we can start with the same expression:

$$\frac{dQ}{dt} = \frac{T_g - T_{other}}{R} \quad \text{Eq. 75}$$

We know that the sinusoidal case is complex: it both requires a guess for the polytropic index and it requires an integral solver. So we'll be approximating with the linear case, regardless of the actual temperature vs time function in the system. So, once you plug that into the heat transfer equation and integrate, you get the following:

$$Q = \frac{T_0 t_p}{R} \left(\frac{2V_f - V_0 - V_f VR^{n-1} - n(V_f - V_0)}{(V_f - V_0)(n-2)} \right) \quad \text{Eq. 76}$$

Since this is a function of n in the exponents and in other ways, this expression makes the equation impossible to solve for the polytropic index. Thus, we'll have to approximate some of the n terms with constants to make it a solvable equation. Since we're interested in whether or not the piston will be efficient, the only case we care about is the one where n is close to one, so we'll plug in $n = 1$ for all of the non-exponent n terms.

$$Q = \frac{T_0 t_p}{R} \left(\frac{2V_f - V_0 - V_f VR^{n-1} - (1)(V_f - V_0)}{(V_f - V_0)((1)-2)} \right) \quad \text{Eq. 77}$$

$$Q = \frac{T_0 t_p}{R} \left(\frac{V_f - V_f VR^{n-1}}{V_0 - V_f} \right) \quad \text{Eq. 78}$$

If you divide the numerator and the denominator by V_f , you get

$$Q = \frac{T_0 t_p}{R} \left(\frac{1 - VR^{n-1}}{VR - 1} \right) \quad \text{Eq. 79}$$

Since we're going to cover expansion more explicitly in this case, from now on VR_C will denote the compressive volume ratio where $VR_C > 1$ and VR_E will denote the expansive volume ratio where $VR_E < 1$. Also, any subscript C will denote a quantity associated with compression, and any subscript E will denote a quantity associated with expansion. For the compressive case, assuming heat flow out is positive

$$Q_C = \frac{T_0 t_p}{R} \left(\frac{VR_C^{n-1} - 1}{VR_C - 1} \right) \quad \text{Eq. 80}$$

and for the expansive case, since

$$VR_C = \frac{1}{VR_E} \quad \text{Eq. 81}$$

assuming heat flow in is positive

$$Q_E = \frac{T_0 t_p}{R} \left(\frac{VR_E^{n-1} - 1}{1 - VR_E} \right) \quad \text{Eq. 82}$$

Next, we can plug in these two equations for Q_C and Q_E and solve for the polytropic index in both cases. For compression:

$$P_0 V_0 \ln(VR_C) = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma - 1)} (T_f - T_0) + \frac{T_0 t_p}{R} \left(\frac{VR_C^{n_c-1} - 1}{VR_C - 1} \right) \quad \text{Eq. 83}$$

$$P_0 V_0 \ln(VR_C) = \frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma - 1)} (VR_C^{n_c-1} - 1) + \frac{T_0 t_p}{R} \left(\frac{VR_C^{n_c-1} - 1}{VR_C - 1} \right) \quad \text{Eq. 84}$$

$$P_0 V_0 \ln(VR_C) = \left(\frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma - 1)} + \frac{T_0 t_p}{R(VR_C - 1)} \right) (VR_C^{n_c-1} - 1) \quad \text{Eq. 85}$$

solving this equation for n yields

$$n_C = 1 + \log_{VR_C} \left(\frac{P_0 V_0 \ln(VR_C)}{\frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{T_0 t_p}{R(VR_C-1)}} + 1 \right) \quad \text{Eq. 86}$$

which can be further simplified to

$$n_C = 1 + \log_{VR_C} \left(\frac{(\gamma-1)\ln(VR_C)}{1 + \frac{s}{(VR_C-1)}} + 1 \right) \quad \text{Eq. 87}$$

or

$$n_C = 1 + \log_{VR_C} (\phi_C + 1) \quad \text{Eq. 88}$$

$$\phi_C = \frac{(\gamma-1)\ln(VR_C)}{1 + \frac{s}{(VR_C-1)}} \quad \text{Eq. 89}$$

Next, to solve for the expansion case, we start with

$$P_0 V_0 \ln(VR_E) = \frac{R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (T_f - T_0) + \frac{T_0 t_p}{R} \left(\frac{VR_E^{n_E-1} - 1}{1 - VR_E} \right) \quad \text{Eq. 90}$$

$$P_0 V_0 \ln(VR_E) = \frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma-1)} (VR_E^{n_E-1} - 1) + \frac{T_0 t_p}{R} \left(\frac{VR_E^{n_E-1} - 1}{1 - VR_E} \right) \quad \text{Eq. 91}$$

$$P_0 V_0 \ln(VR_E) = \left(\frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{T_0 t_p}{R(1-VR_E)} \right) (VR_E^{n_E-1} - 1) \quad \text{Eq. 92}$$

Note that each term is now negative, since $\ln(x) < 0$ for $x < 1$, and $x^a - 1 < 0$ for $x < 1$, $a > 1$. We can solve for n_E to get:

$$n_E = 1 + \log_{VR_E} \left(\frac{P_0 V_0 \ln(VR_E)}{\frac{T_0 R_{gas} m_{gas}}{M_{gas}(\gamma-1)} + \frac{T_0 t_p}{R(1-VR_E)}} + 1 \right) \quad \text{Eq. 93}$$

which can be further simplified to

$$n_E = 1 + \log_{VR_E} \left(\frac{(\gamma-1)(1-VR_E)\ln(VR_E)}{S-VR_E+1} + 1 \right) \quad \text{Eq. 94}$$

or

$$n_E = 1 + \log_{VR_E} (\phi_E + 1) \quad \text{Eq. 95}$$

$$\phi_E = \frac{(\gamma-1)\ln(VR_E)}{1 + \frac{S}{(1-VR_E)}} \quad \text{Eq. 96}$$

We can gather a few insights from these results. First, the closer $\log_{VR_E} (\phi_E + 1)$ is to zero, and thus, the closer $\phi_E + 1$ is to one, the more efficient the process will be (since this would be the limit of $n_E \rightarrow 1$). Therefore, the closer ϕ_E is to zero, the more efficient the process will be. A similar thought process can be used for the compression side with the same result: as $\phi_C \rightarrow 0$, $n_C \rightarrow 1$. It's a bit hard to tell outright, but as VR_C increases, ϕ_C gets further from zero, and as VR_E decreases, ϕ_E gets further from zero.

My analysis has a relatively constant term in the numerator: $(\gamma - 1)\ln(VR_E)$.

Since $\gamma - 1$ is a fixed value assuming the same working gas,

Other analyses have found that the Simpson-Qin number, $S = \frac{t_{process}}{t_{thermal}}$ is very important, and my analysis yields the same result. Since ϕ_C and ϕ_E both have the Simpson-Qin number in the denominator, as the thermal time scale of the air and heat exchangers in the piston approach zero, the piston becomes more efficient, and similarly as the process time increases, the piston becomes more efficient.

Moving on to more quantitative measures of efficiency, we can write the compressive and expansive efficiencies as

$$\eta_C = \frac{(n_C-1)\ln(VR_C)}{VR_C^{n_C-1}-1} \quad \text{Eq. 97}$$

$$\eta_E = \frac{VR_E^{n_E-1}-1}{(n_E-1)\ln(VR_E)} \quad \text{Eq. 98}$$

and we can write the round trip efficiency as the following

$$\eta_{RT} = \eta_C \eta_E \quad \text{Eq. 99}$$

$$\eta_{RT} = \frac{(n_C-1)\ln(VR_C)}{VR_C^{n_C-1}-1} \frac{VR_E^{n_E-1}-1}{(n_E-1)\ln(VR_E)} \quad \text{Eq. 100}$$

plugging in what we found for n_C and n_E , we get

$$\eta_{RT} = \frac{(1+\log_{VR_C}(\phi_C+1)-1)\ln(VR_C)}{VR_C^{1+\log_{VR_C}(\phi_C+1)-1}-1} \frac{VR_E^{1+\log_{VR_E}(\phi_E+1)-1}-1}{(1+\log_{VR_E}(\phi_E+1)-1)\ln(VR_E)} \quad \text{Eq. 101}$$

which we can simplify with a few steps to get the following:

$$\eta_{RT} = \frac{(\log_{VR_C}(\phi_C+1))\ln(VR_C)}{VR_C^{\log_{VR_C}(\phi_C+1)}-1} \frac{VR_E^{\log_{VR_E}(\phi_E+1)}-1}{(\log_{VR_E}(\phi_E+1))\ln(VR_E)} \quad \text{Eq. 102}$$

$$\eta_{RT} = \frac{\frac{(\ln(\phi_C+1))\ln(VR_C)}{\ln(VR_C)}}{\phi_C+1-1} \frac{\phi_E+1-1}{\frac{(\ln(\phi_E+1))\ln(VR_E)}{\ln(VR_E)}} \quad \text{Eq. 103}$$

$$\eta_{RT} = \frac{\ln(\phi_C+1)}{\phi_C} \frac{\phi_E}{\ln(\phi_E+1)} \quad \text{Eq. 104}$$

So the final expression is

$$\eta_{RT} = \frac{\phi_E \ln(\phi_C+1)}{\phi_C \ln(\phi_E+1)} \quad \text{Eq. 105}$$

where

$$\phi_C = \frac{(\gamma-1)\ln(VR_C)}{1 + \frac{S}{(VR_C-1)}} \quad \text{Eq. 106}$$

$$\phi_E = \frac{(\gamma-1)\ln(VR_E)}{1 + \frac{S}{(1-VR_E)}} \quad \text{Eq. 107}$$

This final expression for η_{RT} shows us that as VR_C decreases (and thus as VR_E increases), the round trip efficiency increases. It also shows us that as S increases, the round trip efficiency increases. Finally, it shows us that as γ decreases, the round trip efficiency increases.

It can be shown as well that

$$\eta_C = \frac{\ln(\phi_C+1)}{\phi_C} \quad \text{Eq. 108}$$

$$\eta_E = \frac{\phi_E}{\ln(\phi_E+1)} \quad \text{Eq. 109}$$

and it can be seen that η_E is always greater than η_C .

Thermal Resistance

For the simplified theory, it's all about seeing whether or not a piston will be efficient for the purpose of compressed air energy storage. We will have to estimate an n value to estimate our thermal resistance, and we'll choose an n value based on the efficiency threshold we care about. Since round trip efficiency can be found as follows:

$$\eta_{RT} = \frac{(n_C-1)\ln(VR_C)}{VR_C^{n_C-1}-1} \frac{VR_E^{n_E-1}-1}{(n_E-1)\ln(VR_E)} \quad \text{Eq. 110}$$

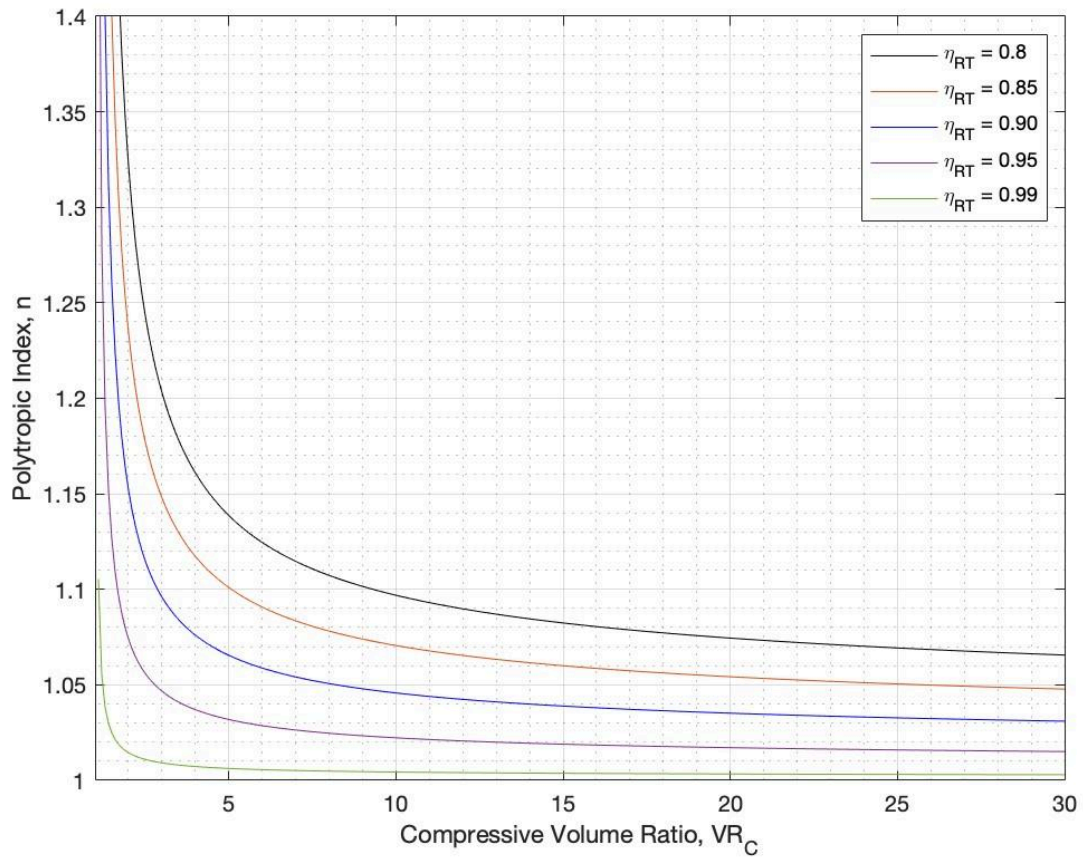
we can assume $n_E \approx n_C = n$ for simplicity, giving the following:

$$\eta_{RT} = \frac{(n-1)\ln(VR_C)}{VR_C^{n-1}-1} \frac{VR_E^{n-1}-1}{(n-1)\ln(VR_E)} \quad \text{Eq. 111}$$

which simplifies to

$$\eta_{RT} = \frac{1-VR_E^{n-1}}{VR_C^{n-1}-1} \quad \text{Eq. 112}$$

We can use a solver to find which n value corresponds to a certain round trip efficiency for different volume ratios. The following graphs show what n values correspond to certain round trip efficiencies for a range of compressive volume ratios.



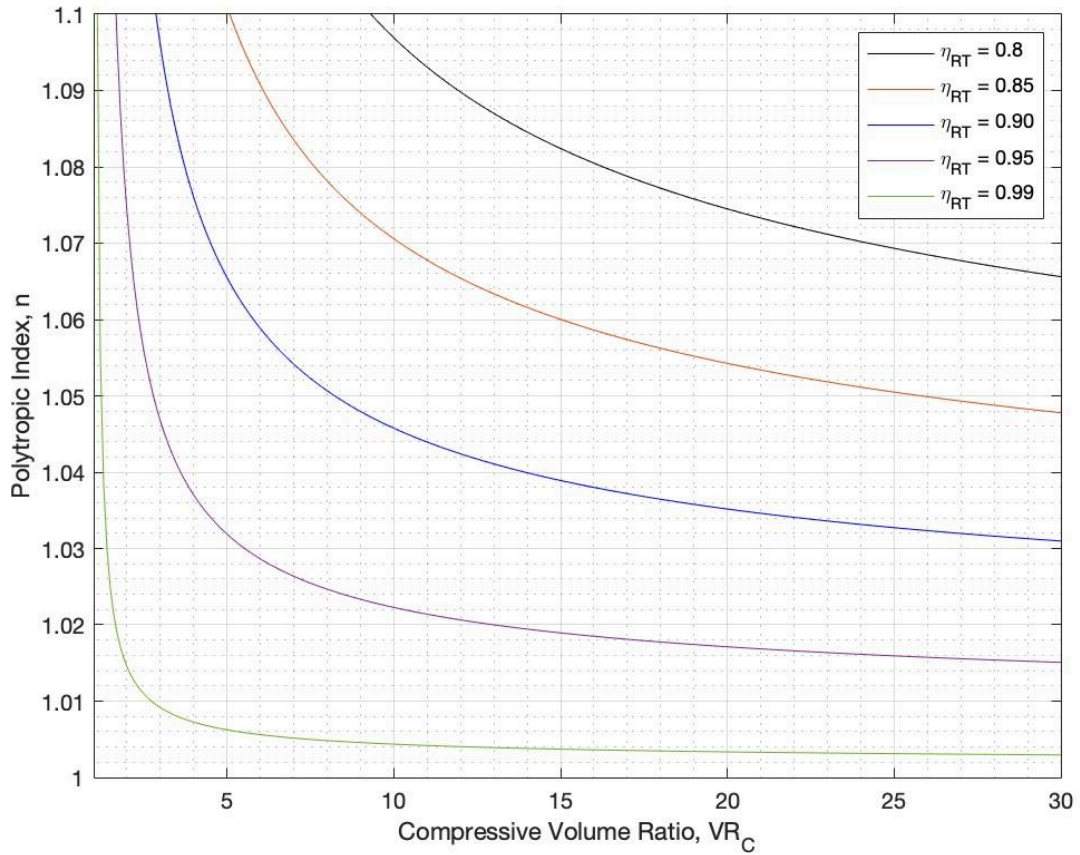


Fig 3 and 4. Polytypic index (n) vs compressive volume ratio ($VR_C = VR_E^{-1}$) corresponding to different round trip efficiencies, full plot and zoomed plot. These were calculated and plotted using Eq. X in MATLAB.

These graphs can be used to inform a guess of the polytypic index for the round trip efficiency threshold you're checking for.

The reason we need a guess for the polytypic index even given the simplified theory is that we have to estimate the Richardson number to determine the convection regime, and therefore we have to know the temperature difference between the gas in the piston and the piston, which is driven by the polytypic index. So, we can find our guess for the polytypic index, n_{guess} , from the above plots given the efficiency threshold we care about. We will then calculate the average Richardson number for our process using the same method as the iterative theory, but instead of the iterative guess, n_0 , we'll use the n_{guess} we got from the plots above.

We'll use the same definitions of diameter-based average Grashof, Reynolds, Richardson, and Rayleigh numbers as before.

$$\overline{Gr}_D = \frac{g(\overline{T}_g - T_s)D^3}{\overline{T}_g \nu^2} \quad \text{Eq. 113}$$

$$\overline{Re}_D = \frac{D\overline{u}}{\nu} \quad \text{Eq. 114}$$

$$\overline{Ri}_D = \frac{\overline{Gr}_D}{\overline{Re}_D^2} \quad \text{Eq. 115}$$

$$\overline{Ra}_D = \overline{Gr}_D \overline{Pr} \quad \text{Eq. 116}$$

$$\overline{Ra}_D^* = \frac{\overline{Gr}_D \overline{Pr}}{\frac{\overline{L}}{D}} \quad \text{Eq. 117}$$

We'll also define all of the other parameters the same way, with the exception of $\overline{T}_g$. Since we are not using complex calculations here, we'll substitute the more accurate integral form

$$\overline{T}_g = \frac{1}{t_p} \int_0^{t_p} T_g(t) dt \quad \text{Eq. 118}$$

with the following simplified form

$$\overline{T}_g = \frac{T_0 + T_f}{2} \quad \text{Eq. 119}$$

to make the calculations easier. Using the polytropic equations we can rewrite this equation as

$$\bar{T}_g = \frac{T_0 + T_0 VR^{n_{guess}-1}}{2} \quad \text{Eq. 120}$$

The term $\frac{\bar{T}_g - T_s}{\bar{T}_g}$ shows up in the Grashof number. Since we know $T_s = T_0$, we can use the above definition of $\bar{T}_g$ to simplify this with the following:

$$\frac{\bar{T}_g - T_s}{\bar{T}_g} = \frac{\frac{T_0 + T_0 VR^{n_{guess}-1}}{2} - T_0}{\frac{T_0 + T_0 VR^{n_{guess}-1}}{2}} = \frac{T_0 + T_0 VR^{n_{guess}-1} - 2T_0}{T_0 + T_0 VR^{n_{guess}-1}} \quad \text{Eq. 121}$$

$$\frac{\bar{T}_g - T_s}{\bar{T}_g} = \frac{VR^{n_{guess}-1} - 1}{VR^{n_{guess}-1} + 1} \quad \text{Eq. 122}$$

We can substitute this into the Grashof number to get the following

$$\overline{Gr}_D = \frac{g(VR^{n_{guess}-1} - 1)D^3}{(VR^{n_{guess}-1} + 1)^{-2} \nu} \quad \text{Eq. 123}$$

The next steps are the same as the iterative version: plug in the values you have to find $\overline{Ri}_D$, and choose the Nusselt number correlation model accordingly. Then, calculate the Nusselt number based on the above models just like the iterative case, and calculate the thermal resistance, R , just like the iterative case.

Final Process

Using the thermal resistance you found above, calculate $t_{thermal}$ and S , and plug in to find the roundtrip efficiency. This methodology uses more approximations and simplifications than the iterative one so interpret its results with caution. It should only be used to get an idea of whether a certain piston will meet your round trip efficiency threshold or not, but it should be easier to calculate than the iterative method.